

# Quang Truong

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## Professional Summary

Doctoral student with 5+ years of experience in software engineering, machine learning, and artificial intelligence. Skilled in designing and deploying scalable ML systems and deep learning solutions. Expertise in **Graph Machine Learning, Relational Deep Learning, Knowledge Graphs**, and **Large Language Models (LLMs)**, with publications in **NeurIPS, AAAI**, and **JMLR**. Strong background in industry research, large-scale model development, and open-source contributions. Experienced across the ML lifecycle, including data processing, model design, training optimization, and distributed computing. Proficient in PyTorch and Python, with solid foundations in algorithms, databases, and high-performance computing (HPC). Seeking a **Research or Engineering Internship (Jan.-Sep. 2026)** to advance applied AI research and deliver high-impact, production-ready solutions.

## Education

### Michigan State University

Ph.D. in Computer Science

East Lansing, MI

Aug. 2024 - May 2028

### Dartmouth College

M.S. in Engineering Sciences

Hanover, NH

Sep. 2022 - Jun. 2024

## Selected Publications

‡ indicates equal contribution.

### Peer-reviewed Publications

- [1] A Pre-training Framework for Relational Data with Information-theoretic Principles  
**Quang Truong**, Zhikai Chen, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2025**  
*NeurIPS'25: The 39th Annual Conference on Neural Information Processing Systems*.
- [2] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains  
Mustafa Hajij‡, Mathilde Papillon‡, Florian Frantzen‡, ... **Quang Truong**, and ... Nina Miolane. **2024a**  
*The Journal of Machine Learning Research (Volume 25)*.
- [3] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes  
**Quang Truong** and Peter Chin. **2024b**  
*AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*.
- [4] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification  
**Quang Truong** and Bo Mei. **2021**  
*ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*.

## Open-source Projects

### TopoX

- Provides reliable and user-friendly building blocks for computing and machine learning on topological domains that extend graphs.
- Adopted open-source software engineering best practices with comprehensive testing, documentation, and CI/CD pipelines.
- Starred 500+ times across 3 different libraries on GitHub.
- Published at the Journal of Machine Learning Research (JMLR).

## Skills

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|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming             | Python (Expert), C/C++ (Advanced), Java (Proficient), R (Proficient), MATLAB (Proficient)                                                                                 |
| Machine Learning        | Graph Machine Learning, Relational Deep Learning, Knowledge Graphs, Recommender System, Large Language Models, Computer Vision, Data Mining                               |
| ML Frameworks           | Pytorch, Pytorch Geometric, Pytorch Frame, NetworkX, TopoX, Pytorch Lightning, LangChain, Hugging Face, Scikit-learn, Matplotlib, Numpy, Pandas                           |
| DevOps & Infrastructure | WandB, Docker, Singularity, Pytest, Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, CI/CD, Monitoring, Distributed Systems                            |
| Software Engineering    | Spring, Agile, JIRA, SCRUM, Git, Linux/Unix, MongoDB, DuckDB, MySQL, Unit Testing, Performance Optimization, Automation, Algorithm Design, Data Structures, System Design |

## Research Experience

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### DSE Lab, Michigan State University

East Lansing, MI

Doctoral Research Assistant (Advisor: **Jiliang Tang**)

Aug. 2024 - Present

- Develop scalable transformers and characterize necessary conditions for the models to be strong Link Predictors (**Snap Research Collaboration**).
- Design and implement distributed training pipelines for large-scale graphs, optimizing performance across multiple GPUs and ensuring scalability for enterprise-scale data.
- Create automated testing frameworks and continuous integration pipelines to ensure code quality and system reliability across research projects.
- Develop specialized pre-training procedures for graph machine learning on relational databases, thereby enhancing predictive modeling for node-level classification, link prediction, and temporal forecasting tasks (**Snap Research Collaboration**).

### LISP Lab, Dartmouth College

Hanover, NH

Graduate Research Assistant (Advisor: **Peter Chin**)

Sep. 2022 - Jun. 2024

- Introduced a novel approach to Topological Deep Learning (TDL) leveraging simple paths, significantly enhancing graph expressivity and surpassing the theoretical limitations of the 3-WL test.
- Developed Path Isomorphism Network, a state-of-the-art TDL model, which delivered outstanding performance on benchmark datasets: PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).
- Developed an anomaly detection pipeline for human behavior analysis using geospatially-tagged data, as part of the **IARPA-HAYSTAC** funded project.

### Vision Lab, University of Illinois at Chicago

Remote

Research Intern (Advisor: **Wei Tang**)

May 2021 - Aug. 2021

- Designed a context-aware 3D object detection architecture that captures both object-wise and object-context relationships, improving spatial understanding.
- Boosted model performance through cross-modality alignment, improving the feature representation of objects.

### ML Lab, Texas Christian University

Fort Worth, TX

Undergraduate Research Assistant (Advisor: **Bo Mei**)

May 2019 - May 2021

- Developed an advanced Vehicle Re-identification pipeline utilizing Generative Adversarial Networks (GANs) for adaptive domain learning, resulting in a 10% improvement in mean Average Precision (mAP) over the baseline.
- Engineered a novel filtering algorithm that reduces the training data requirement by 25% while maintaining model performance, significantly improving computational efficiency.

## Presentations

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### Invited Presentations

- Topologically-driven Neural Networks, *HPCC Special Seminar: Decoding Computational Challenges, the Advanced Institute of Interdisciplinary Science and Technology - HCMUT, VNU-HCM*. **2024**

### Conference Presentations

- Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes, *AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*. **2024**
- Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-identification, *ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*. **2021**

## Professional Services

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- Conference Reviewer, *Learning on Graphs Conference (LoG)*. **Fall 2025**
- Program Committee Member/Conference Reviewer, *Annual AAAI Conference on Artificial Intelligence (AAAI)*. **Fall 2025**
- Conference Reviewer, *Conference on Neural Information Processing Systems (NeurIPS)*. **Summer 2025**
- Journal Reviewer, *IEEE Transactions on Big Data (TBD)*. **Summer 2025**
- Conference Reviewer, *International Conference on Learning Representations (ICLR)*. **Fall 2024**
- Journal Reviewer, *ACM Transactions on Knowledge Discovery from Data (TKDD)*. **Fall 2024**

## Honors

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- NeurIPS 2025 Scholar Award, *Neural Information Processing Foundation*. **2025**
- VEF Fellow, *VEF 2.0 Program, VEF Fellows and Scholars Association*. **2021**
- Vingroup Scholarship Recipient, *Master's & Ph.D. Scholarship Program, VinUniversity*. **2021**
- Bronze Medal, *ACM-ICPC South Central USA Regional Contest [URL]*. **2019**